Interpretation of Measures of Central Tendency

Statistical Summary of Academic Performance and Salary

This assignment interprets the mean, median, and mode values shown in the given table. The serial-number column (sl_no) is excluded because it is an identifier and does not represent a meaningful academic or salary measurement.

Dataset file name: Placement.csv

1. Meaning of the Three Measures

Mean: The mean is the arithmetic average. It is calculated by adding all observations in a column and dividing the total by the number of observations. It gives an overall idea of the typical level of the data, but it can be affected by unusually high or low values.

Median: The median is the middle value when all observations are arranged in ascending order. If there is an even number of observations, it is the average of the two middle values. The median is generally less affected by extreme values than the mean.

Mode: The mode is the value that occurs most frequently in a column. It identifies the most common observed value. A dataset can have more than one mode if multiple values occur with the same highest frequency.

2. Values from the Given Table

4]:

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186	288655.405405
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0

3. Step-by-Step Interpretation of Each Variable

3.1 SSC Percentage (ssc_p)

- Mean = 67.30. The average SSC percentage in the dataset is approximately 67.30%.
- Median = 67. The middle SSC percentage is 67%. This means about half of the observations are at or below 67% and about half are at or above 67%, subject to repeated values.
- Mode = 62. The most frequently occurring SSC percentage is 62%.

- Interpretation: The mean and median are very close (67.30 and 67), suggesting that the centre of the SSC percentage distribution is around 67%. The most common value is somewhat lower, at 62%. These three values alone do not show the full spread of marks.

3.2 HSC Percentage (hsc_p)

- Mean = 66.33. The average HSC percentage is approximately 66.33%.
- Median = 65. The middle HSC percentage is 65%.
- Mode = 63. The most frequently occurring HSC percentage is 63%.
- Interpretation: The mean is slightly higher than both the median and the mode. This indicates that the average is being pulled somewhat above the most common and middle values. A few higher percentages could contribute to this difference, but the table alone cannot confirm the shape of the distribution.

3.3 Degree Percentage (degree_p)

- Mean = 66.37. The average degree percentage is approximately 66.37%.
- Median = 66. The middle degree percentage is 66%.
- Mode = 65. The most frequently occurring degree percentage is 65%.
- Interpretation: The mean, median, and mode are close to one another. This suggests that the central values of degree percentages are grouped around the mid-60s. It does not mean that every student scored near these values; individual observations may vary.

3.4 Employability Test Percentage (etest_p)

- Mean = 72.10. The average employability-test percentage is approximately 72.10%, the highest mean among the percentage variables shown.
- Median = 71. The middle employability-test percentage is 71%.
- Mode = 60. The most frequently occurring employability-test percentage is 60%.
- Interpretation: The average and median are above the mode. This means that 60% is the most common individual score, while the centre of the data based on the mean and median is higher. Higher scores may be lifting the average, but further analysis is needed to determine the distribution shape.

3.5 MBA Percentage (mba_p)

- Mean = 62.28. The average MBA percentage is approximately 62.28%.
- Median = 62. The middle MBA percentage is 62%.
- Mode = 56.7. The most frequently occurring MBA percentage is 56.7%.
- Interpretation: The mean and median are almost equal, while the mode is lower. The typical central value is around 62%, but 56.7% is the most frequently observed value. The mean being slightly above the median may indicate some influence from higher observations; this cannot be established conclusively from these three measures alone.

3.6 Salary

- Mean = 288,655.41. The average salary represented in the dataset is approximately 288,655.41 in the dataset's salary units.
- Median = 265,000. The middle salary is 265,000. Approximately half of the salary observations are at or below this value and half are at or above it, allowing for ties.
- Mode = 300,000. The most frequently occurring salary is 300,000.
- Interpretation: The mean salary is higher than the median by approximately 23,655.41. This difference may reflect some higher salaries pulling the average upward. The most common salary is

300,000, which is above the median. The currency and salary period (for example, annual or monthly) are not specified in the table, so they should not be assumed.

4. Comparison Across the Variables

- Highest mean among the percentage variables: The employability-test percentage has the highest mean at approximately 72.10%.
- Lowest mean among the percentage variables: The MBA percentage has the lowest mean at approximately 62.28%.
- SSC and degree percentages: Their means are very similar, at approximately 67.30% and 66.37%, respectively.
- Mean and median are close for SSC, degree, and MBA percentages. This shows that their mean and middle values are similar, although it does not by itself prove that the distributions are symmetric.
- The mode is below the mean for every percentage variable shown. The most frequently occurring value is therefore lower than the average in each of these columns.
- Salary has a mean of about 288,655.41, a median of 265,000, and a mode of 300,000. The difference between mean and median suggests that the average may be influenced by higher salary observations.

5. Conclusion

In conclusion, the measures of central tendency provide a simple summary of academic percentages and salary. The mean represents the average, the median represents the middle value, and the mode represents the most frequent value. The employability-test percentage has the highest mean among the academic percentage columns, and the MBA percentage has the lowest. The salary mean is higher than its median, which may indicate that higher salary values influence the average. These findings describe the central values of the data, while a more complete understanding would require measures of spread and visual analysis.